[00:00] 发言人 1: 是的，爱工作啊，所以是那个。Uh，that is making even more，um，obvious，there guys。That means about one sheet sheet alright we can do this like write up even like this on on all five of the the april paper and bring it into the into the world with ah nature you see what I mean.

[00:28] 发言人 1: 对呢，but just one，all right，just one paper，you can bring all all size，i can’t bring more than one，all right。yeah，so this is um. Um，the policy for big term，to just make sure，like，if you have any friends，you can just remind them about，um，like this policy as well。

[00:53] 发言人 1: Um，I’m not going to like post it on canvas，because I’m not sure if like this is allowed。Um。By college or something，but I think it’s okay because it’s sort of like it’s a half open book exam.

[01:06] 发言人 1: It’s not like totally close for exam or make them I don’t want make you feel so bad about it. And the other thing is. Yeah，like just make sure you are familiar enough. With a let me see with a materials or some distributions such as um a song distribution um um the nori distribution by normal distribution helping distribution and flashing distribution alright i used to be familiar with this five different distributions which will be very useful.

[01:48] 发言人 1: 啊你你没事。Alright，I think this one，I mean information I can provide for the lecture，and then I can start with today’s um material。啊，嗯，再来一个，嗯，啊，One more information that I can talk about is，ah，just in case make sure that this is correct，and provide false information，like，hm，see.

[02:34] 发言人 1: Oh yeah， so basically classroom is not in the right of maker， so that means if you want to raise in this lecture， you can still have a good raising in them。Right，Uh，which can easier very little bit，and inside most of the the things except in the in the nature are they，Uh。Information that we learn for the first three courses，the first three lectures.

[03:08] 发言人 1: I believe，alright? There might be a little bit information of our current file. You just need to know that you just need to make sure like you have a little bit information about this this two lectures.

[03:20] 发言人 1: Most important part is like the first three weeks. Alright，but first three weeks，which can associate with one of the open like one of the open answer questions in the vector。I think this is like very important information for you to like prepare for the course.

[03:45] 发言人 1: So you’re gonna start with that channel. So today we gonna mention like about local estimation of classroom，which is a really interesting part in the shell learning. So the learning objective today is um uh this five points the first is fine explain why the case um distribution down in1826 second as everyone does define and local um contribution from here.

[04:18] 发言人 1: Third is like explain how environment right changes more medicine of variability and forth is the right membership public which is one of the classroom method can introduce today um from the estimated distance and finally compare it means um doctrine make sure the membership. So we’re gonna start with local estimates. Suppose we observe ten data points，x one to x ten.

[04:48] 发言人 1: All right, a parametric model that we basically learned last time, like, hey, this is a polishing model. They have like several parameters. We have to estimate this model is fine.

[05:04] 发言人 1: The whole data set using a big distribution family. For example, for ocean right, it only uses two different parameters. You know, it must be. Or oceanic model now we learn last time we have ik that is say um the proportion or the ratio of different um oceans and then you can see my square are they making that families of um this oceans. Some more data input the prime estimates are the basic model form in this space.

[05:45] 发言人 1: Now inside where is like where they um you know like assume how the data can be distributed using a model right data and basically like we the only thing that we um we are flexible with. In this parameter models are the parameters themselves, right? Right, you know, like whoever is doing that window structures, so we use either ocean or the mixture of oceans to estimate the distribution of the data.

[06:15] 发言人 1: And basically, it’s just changing the parameters of each ocean or you know, like something like that. So local methods that that’s how the methods we’re gonna introduce today. All right, local methods instead that observe neighborhood is targeting the destination.

[06:33] 发言人 1: All right, then if we look at the data points nearby, like uh for example neighborhood or um uh a threshold with uh like a distant threshold. 对他们应该更擅长，所以这是。Um，so first one，introduce was local，but like local data consultation。Consider location that’s where we want estimated.

[07:00] 发言人 1: Um，that’s，and should say，or like never really is，really is is a distance，right? It’s a distance from this point，that，um，you know，like inside this distances，or outside distances，are they，um，you know，like the their points，close to this point，or like apart from this point。So just a number of radius R larger than zero, and an observation Xi is considered nearby when the Xi distance to X is smaller or equal to R.

[07:34] 发言人 1: All right, these are explicitly you should use like the circle with the radius R, right? Um，and the local neighborhood is therefore reacts like trying to find all the x eyes。那 fall into 啊 neighborhood of ice which basically is here your base and smaller equals to here.

[08:00] 发言人 1: Like for R, right? So the case if you have the text, and like a specific component of anything, see. 啊。And actually, this is the X.

[08:15] 发言人 1: All right, no, I just basically drawing a circle. It’s not a circle, but you know, sort of circle. And then like distances are right.

[08:25] 发言人 1: All the points inside this circle will be considered. The neighbor will ask. 哎，这个是啊，还可以啊，啊，And the NRS is transforming set that containing all the neighbor points of us.

[08:50] 发言人 1: And and we we let they can uh not being like this uh like the the uh the capital and our acts to be the. Or the elements inside this set, right? It’s the count of elements of inside the set.

[09:07] 发言人 1: So this is the number of, you know, basically the number of points inside this network. But then for the same radius R, a larger and indicates greater concentration. This is the count of time because it is exactly means like there are more points.

[09:25] 发言人 1: Inside this um big circle, right? So one dimension they never put one access on x minus r x plus r because one dimension have a certain one dimension. Nothing thing you have is a like a segment right? This is a segment.

[09:43] 发言人 1: The total width is um two r and the probability is nice inside this interval as i could take them in innovation of the. Or policy can get from x minus r to x plus r, but the peak here could be down. 嗯，或者是一些嗯，just the amount，and the interval is small enough，now，if you change the place of what as maybe，but they can have，if you approximate equals x，equal to p x。

[10:15] 发言人 1: Right, but then in this case, if we say like for this whole interval, right, from next x like minus r to x plus r, you have almost the same, you know, like models to use. So the thing here is really, let’s just draw a simple graph here. This is even easier than the graph on the.

[10:36] 发言人 1: Right，order，so if this is as。And r is like small enough is x minus r。And this gonna be x plus r, right? And the density function is looking like.

[10:57] 发言人 1: Something like this, alright? And then what it says is like repeat x. this is ps right this is ps. And if in this interval, if r is smaller enough, this whole interval can be estimated using the same y or using the same dx.

[11:20] 发言人 1: So basically it’s like the value of the function of function value of the dx here can be approximately equal to p u and they can basically write it as they rectangular area, right? Because rectangle has they uh the two sides one side to be two R and other side to be uh P X so you can approximate equal equal to two R multiply by P X. Alright, so this is like a thing I’m introducing here.

[11:56] 发言人 1: 啊，你先 local power was estimated。After some as well，so more observations，and our acts are all inside the interval，has an entire for progress masses，on the number of comes inside the interval over all of the costs，all right, like the number of experiments or the number of observations，combined it with, you know，like the function that’s up here。Right, you have yes like this, and then you can get they, um, you know like the integration or the the the probability approximately equals two R multiplied by PX, right? This is approximation, this is also approximation.

[12:44] 发言人 1: If you are equaling this to optimize like approximations, you get the PX. 呃，people，呃，m r x over two m r，and this is basically the local sample fraction over local best。And the PS here, the PRX here, R is like the radius, and here means this is estimation.

[13:09] 发言人 1: So this is a estimation that we just estimated using you know like the radius, sorry, rectangle area to approximate the probability from X minus R to X plus R. And also we have, you know, like empirical estimation like n r x over n because n r x is a number of tons of observations on into this, like on into this segment. So this can be equal to each other and make an guess the estimation of density.

[13:47] 发言人 1: Alright，today is how we use local consoles to provide。And if you use this formula, um, alone, you can say, um, for the university here. As some local sample fraction, sorry, it’s not the emergence about an r x over m is local sample fraction and over the local but local but is for arm right? And it’s the histogram applies the same idea using our experiences that h larger than zero be the interface.

[14:22] 发言人 1: And let’s MVX, the number of operations in the big containing X, the empirical probability mass of that being as MVX over X, and dividing by the input is the density of MVX over NH. The only difference here is that we are changing. They are the radius insulin H right because two R two R equals H and like H here means the width of the of the beam which is is two R this is the only difference alright so the estimate.

[15:00] 发言人 1: constant 说一马一拳。嗯，呃，English is certainly amount of，呃，local aspiration。For a small age, we use few points per being, and if if means like we have more details and more and the larger H more points per being means like more small less details。

[15:26] 发言人 1: right it basically controls the range of the the like the red. You want you want an operation with right that is it controls resolution high resolution okay 啊你可以问问，same rule related to play by day，但是你可能可以问问，right? So uh this is an example of the history grand on great facts even you have the same in web even you have.

[16:00] 发言人 1: The same H alright but you can still have different history if you choose like different initial point initial um um or like the central point you can see that in this example right should be the boundaries and change um they the comfort each thing. Because for each thing, you are just like doing statistics with how many counts fall inside that thing, right? If you move it, if you move it a little bit, if you shift the boundaries, you can have different counts for each thing, and therefore you can have different, you know, like statistics around this two different around the same.

[16:43] 发言人 1: 啊，这是优势。来。so food relations of the first is continuous jumps between beans this this is the first indication of this sort of is brand and second is.

[17:00] 发言人 1: Sensitivity is great placement. If you have different grid for different grid points, the corresponding histogram is gonna change, right? So this is the reason why we all want to use histogram.

[17:12] 发言人 1: We all want to use like this, um, very simple interpretation of um, of intensity. It’s too rough. This thing is too rough, and it’s this great. Right, it’s this way. So we want to do is we use continuous, you know, like continuous estimation. So fix the discontinuity of this sort of jumps of his brain, and also you don’t need to care about like the great placement because if you have the best things that the mass function, you can do whatever integration has. appropriate，so maybe or terrible does the estimation，and we gonna like to reduce the management，the most basic in honor is by sending a school，um neighborhood，on the house of relation。这就是difference，所以这里我们用 KD 或者 competency estimation。

[18:13] 发言人 1: So Instagram rules，um，like observations and experience，as we just introduced，a terminality estimation of K D E，places a local contribution along each operate observation，and an average is some。Alright，but each point you can see on the like the first row，we have like seven different distributions。And we can just average up with different weights，alright，with this sum up to one，and then you can basically have the average，let’s see on the comment like the。Floor side right you can see operate operation of we can have like this um average or integrated aggregated um.

[19:00] 发言人 1: That’s the estimation, which uh which is continuous, and we can visit for each um you know for each uh segments, for example from zero to one, you can just um you know like calculating the area under it, and then we don’t have any problems with um you know like the value is changing or something like that. So you see those are like the points. We have for the observations the yellow box and each of the response to one of distribution here after averaging with different ways and ways can be learned.

[19:39] 发言人 1: All right, ways can be learned. So with a smooth terminal, there are observations where overlying contributions and a smooth as they heard. This is a man here on.

[19:52] 发言人 1: K B E，so we have cooperation X I contribute say，um，a function，recall Q I H X，H means like the progress of the。Yeah, like reservation and I need to have contribution or I get a chance and equals one or eight with pay function pay is kernel that you can turn into the Gaussian kernel or even some other kernels Gaussian is almost. 啊啊，Most common kernel。

[20:26] 发言人 1: 或者可以。So let’s go have the pay here. Um, The thing I got you the case x minus x i for h.

[20:34] 发言人 1: All right. Um，The current case determines the shape，the shape of the the uh distribution，right? For example，dancing is fashionable，right? If you have some other sugars，my interests，whatever，it depends on the which distribution you’re using for the K function，and the plant that age，the current is the best of the estimation，right? The range。

[21:00] 发言人 1: Uh，into the consideration，or compared to the corresponding H, how far it acts，um，like axis，um，with acts，right? Is that like two times of H harder，or it’s like0..1 H harder，right? The closer the point is，the smaller the fraction will be。Right, so basically we can say that after one of the height to preserve area, right? And the constant x minus x i over h measures this does from x i in fact, but units for cultural kernel increasing h makes contribution wider and lower.

[21:44] 发言人 1: This is also the same thing that we well for all changes distribution all the time, right? You have a larger. 哦，这款是哎，你那个啊，当时就是 why 那么高。So here’s what I’m asking, as you may can see using face games again. Limitations are on the result and on the locations because we already see that if you move, if you shift the boundaries of the case ground a little bit, you can change the whole distribution right? And this is changes abruptly.

[22:22] 发言人 1: As um being honors，that means there are a lot of jumps，these are like discontinuous or um like this from estimates，isn’t this? Um，so that’s the reason why I want like change the this great um problem of runs and mass function into a um probability density function using the first estimation。啊，我这是看那些 nearby alterations，third education are，nearby alterations are treated as explanations like one game。

[22:57] 发言人 1: One of that things and inside the behind. Have very close obligations, if even they have different values. All right, one is like3..1, yeah,3..2.

[23:09] 发言人 1: Even they have different values, they are treated the same if they are all into3..4 like this game. Already have no differences. We don’t know like how close or far to this point with the other points in the data set.

[23:24] 发言人 1: Right, so this this are like the limitations of hasten grounds have. So that’s that’s also motivate more on using permanent estimation. Right, maybe gives the idea of local counting, using the uh that age changes the neighborhood. Right，we we we are basically changing from this beans to uh small neighbors around operations，or we can say maybe replaces heart beans with uh small contributions from individual data points。

[24:00] 发言人 1: Uh this idea is actually really common in machine learning and you see the syllabus for example for SVM right uh we also gonna introduce what is SVM then uh why we use soft margin for SVM or some other curvature regularization or something like that always trying to use um you know some soft or smooth. components that make sure the hard materials are hard restrictions to be observed. Otherwise, it’s really hard for this model to be flexible for data set for different data sets.

[24:45] 发言人 1: Uh，and then what I want to say is like issue observation and creates uh one density contributions，suppose we have one data on X I，instead of putting all is probability mass into uh one team。Uh，maybe actually，um，surprise it smoothly along icon，and contributing like this，as we just mentioned，right? Eye size after points and access location，where density is evaluated，it should say neighborhood well，and cares the shape of local contribution。

[25:19] 发言人 1: As a defining kernel, which is yeah, we should talk kernel, and each automation is one local distribution. And it is that all personal has like code of every passport and operations. And the KPI basically just averages the contribution from each automation. We have all operations are not together, and just by in by hand.

[25:43] 发言人 1: Alright，or equivalently，um，we can even take one of the eight outside the uh summation，because each of this distribution is uh you know normalized with one of the eight，so we can take that out，in which case we have the。The probability that are sorry，the density function that we estimated is equals to one over n h multiply by the summation of the currents。alright!

[26:13] 发言人 1: For each single x alright so the answer is continuous so for each single x you can compute the distance from this x to um um all the x size over a and change changes distances into um you know or transform it using a function and

then like. 啊。being average by one over mh，alright。So the distance of x and each data point x on on access all right is determined by all your vital relations.

[26:54] 发言人 1: And currently this time is all observation what’s this nearby locations now it carries. of all the numbers when zero really if if you have and smaller than is not an current function right and the of the the of the kernel along equals. Therefore one kernel represents a probability distribution with total mass one and how like common choices as we talk like um mention is pushing her right is effectively give us a smooth contribution as we mentioned or box kernel.

[27:42] 发言人 1: Box kernel is like, um, um, yeah, it is doing on distribution inside a specific box, right? If you have box to be done, like the the right of the um distribution is H, then you have all. Points inside this interval can be having this the one of their h this basically ensures you know the k u is equal to zero which is one of h h is not equal to zero so one of h is also not equal to zero right and also the integration of this part is the right time so right time can be like the air after right time can.

[28:24] 发言人 1: Calculate it has a direct usable which is h multiply by the height which is one of eight. This gives you one right? So you can either use bouncing kernel or box kernel.

[28:37] 发言人 1: The first one for sure is some, you know, like fragmented because it’s most. And the kernel controls the shape, but usually it has less influence than the flat and thread. 你搬过来我高选的是啊。you know H right.

[28:57] 发言人 1: If you have a larger age, it get your. Why the bomb? And it was like a bomb seems like you know like a small like a I would say like a house.

[29:13] 发言人 1: He was great right like a power a bomb on a home roll. So all things like that. Like this, and this size is very high so.

[29:24] 发言人 1: 啊，这是的话，啊，你 still on the road，this is the road，啊，sometimes you have to like。Grace your great because you see some palm song flow. This like palm song, all right? This like palm song.

[29:40] 发言人 1: This is the English course instead of a chamber in course. Um, anyways, uh, let’s back to the definition of um processing here. So why does KBE who tends to pipe to one of the age? So this is what we call it back right?

[30:00] 发言人 1: Each contribution, a larger age give you a wider bomb, alright? And then spread the ocean distribution with wider and lower high, and a smaller smaller age give you a narrower bomb, which means the high is higher, and distribution is more like a peak, right? You have the equal media.

[30:24] 发言人 1: And changing the web should not change the total probability mass for one operation. How is like this? And if you use that different age, age is going to give you the same result because of this.

[30:39] 发言人 1: All right, if you have beautiful x minus x i over h and want to like an operation of this part. Alright，uh，for from this，you can survive like DX equals um H over DUR，H multiply by DUS。So if you have like an equation of this。If you want to H X, um DX in the original office, if you plot the H X by this part, I’m not going to

that yet, but like at least you can have it as K to the U, right? K to the U, I’m going to keep going to one because, um, Sorry, I’m gonna stay here.

[31:22] 发言人 1: Basically, you first find this orders, and you have, you know, the k equals a u here, let’s change the b s into h d u, and h can multiply by one over a, and this from cancel, and then you have like the integration u equals one, this is some. And it does the amount of this take five this by two problems the first is five for each data point for every data point on the x axis the corresponding that the cycle equal to zero how you access all right no. Because each component of the density that contributes from each observation are positive, and each one has a weight one over n.

[32:11] 发言人 1: Sorry, one over a or one over n, one over n, right? So they have one over n as their weight. Think about like all the numbers are positive or nonnegative multiplied by a nonnegative rate.

[32:25] 发言人 1: Like the product, the sum of the products still are positive or not active right? And the integration of this all way from one because if you sound like if you put these estimation of each distribution and little one and outside you can see like each of this. 啊，integration at one，and the sun and times is ten，right，one multiply by that equals one，so this is.

[32:55] 发言人 1: Reason why is also about us for us to ask them all right. And this is not a good, it’s just an estimate. All right, the heavy value pHX, pHX or pHX has, and the density high, not a probability. And probability requires expensive function, or both expensive function. All right, the probability requires an interval, or from A to B, the probability equal in so.

[33:28] 发言人 1: The integration from a to b of the p x p h x times dx alright for small interval for small interval from x to x plus delta delta is a very small shift and then corresponding of the probability that is high power from. 啊，毕业太惨，mostly play by this，嗯，down，therefore density value can be larger than one，it’s probably normal，嗯，people adoption，right，you have adoption，right，adoption，right，you have。为什么啊。sigma or more areas you have a very um like a p t ocean right.

[34:10] 发言人 1: So therefore, sorry, select method reason why this back in the last one, okay? Some narrow ocean map, make it make a high peak, high peak, plus you have total everyone, this is something like this. Outline the summary ocean kernel is looking like this.

[34:30] 发言人 1: This is how we get use k function, k function, k function will be one work. Uh, square of two time multiplied by the exponential of what a maximum use square over two. The use square as if you take the original view here, which is x minus x i over h.

[34:51] 发言人 1: So the use square will be like x minus x i square over two h squares. All right, if you if you just like provide h of two. This is an option this is the main x i the sensitivity age match um there are operations um um country of move to uh more to the uh estimating at x this operations country of less and their contributions still remain positive.

[35:21] 发言人 1: This is the main idea of kernel estimation. And if we give a memory example for box kernel example, the box kernel as we said is basically a uniform distribution over a small interval. So if you have a u right, the one one half multiplied by u.

[35:43] 发言人 1: Uh，If like the indicator function，that means if the the like the criterion here sintax box，no，you’re gonna say what this equals one，all right，if the criterion is not sintax box，no，we say this equals。Zero，so this is like just an indicator of if the the distance the practice is strong，if the the distance or if the u is smaller or equal to one。We’re gonna have this like I equals zero, otherwise equals one, otherwise zero.

[36:18] 发言人 1: So this A has a simple containing interpretation. Right，if you have E H X has equal to the number like the hashtag，it means like the number of right, the number of I where X minus X I smaller with H。This is a count of like Xi that is also packs within the distance H and then also like over two and two and H this is a problem that we have to work right and usually counts.

[37:00] 发言人 1: Observations inside um numer this here right numer constant observations inside this um domain are sustainable and the denominator is um sample size times um one over that alright this is sample size. The end here and the window width is two h right is two h to consider x zero x one equals zero x two equals one h equals one there are two contributions overlap over the usable zero and one zero to one right and in this case. 一会儿那个是啊。

[37:44] 发言人 1: location x next zero point five zero point five one point five to three observations in the one zone with next zero point five we have one right and for zero point five we have two like this is the only point we have two. Observations and one point five has one because x equals one which has you know like also distance and three has nothing inside the box because there’s no points from the super port right there’s no points from the super port so at x equals zero point five all observations contribute. And this case，the B one half equals zero point five equals one over two, which is a rational and one over two，cos one over two，and this is us one over two。

[38:40] 发言人 1: The estimate is higher where more local contributions overlap。alright. So this is idea of it this is idea of honey 呢嗯 you know ecstatic local best. 嗯嗯。

[39:00] 发言人 1: See estimates across different but you can get very different results as you see here right the same operations can produce very different estimates if you have very small h. Right, like this. It gonna give you like you can see where the data points are distributed by this yellow bars.

[39:20] 发言人 1: And it gonna give you like very separated pieces. Right, very separated pieces. Maybe sometimes it’s fine, but um you know like in general it’s more this little too small for they uh selection.

[39:35] 发言人 1: Okay, if you wanna have a useful detail, right? Maybe just like increase it a little bit, you gonna be more strong. It’s more small, and if you have, if you change it too large, you really can’t see anything, right?

[39:50] 发言人 1: This is not really telling us any information about the plane distribution because you’re going to see that there’s large empty area. right here and empty area here there’s no signals. On this curve, right? There is no signal on this curve. So this time how you gonna choose they uh the hardware, if you have select on it, and then see the shape of a uh positive option, then decide which is uh the best age.

[40:23] 发言人 1: And this can also extend this most dimensional for for calculations estimates observations x one x ten a scalar that what k is on zero like work so basically this on x actually is not all right it has like one bottom value which is previous example i only consider x to be a number. And now we are thinking like these are different of operations with um you know more than one entries for each one it can be like a one dimension vector one dimension vector. So say that together like this alright and the k here is the dimension of density right we have this for each of the um you know like the dimension have the integration one to one scaling temperature um 应该采用啊，company by h multiplies following by h p。And the actor one of the age of power the preserves um the area or volume honor each contribution.

[41:32] 发言人 1: And as social cake，呃，washing kernel is，呃，multi dimensional washing as well，right? Um，yeah，observation，I saw contribute，呃，you know，you can just say，呃，part here，okay，you can survive each function，each contribution has in Xi，and various each square I think。which means we are basically saying that you know all this different um.

[42:00] 发言人 1: You know, like dimensions are independent with each other, right? Because we only have a backlund trees and they have the same values squared. So shared functions like uses the same instance now, you never comment features out here for that.

[42:21] 发言人 1: This means we have to do normalization of word to do the K E estimation, right? Because think about we already assume like the variance of each dimension to be the same. If I have two features, one is for example your age, alright? The other one is your weight.

[42:48] 发言人 1: So for ages, I think like most of you guys should be on twenty-one to twenty-three, right? And for the so basically everyone is inside this range. And we assume like there’s a variance, but for our weight, you can range from like40 kilograms to even100 kilograms.

[43:10] 发言人 1: Right? This is a larger range. Right? This is definitely a larger range. So what we want to do before we use this model is we have to, we’re going to scale each of the variables into, for example,0 to1.

[43:24] 发言人 1: So you got100 one, you’re gonna be scale zero, you got300 scale one, and you got like4 kilograms, you got scale zero, you got100 kilograms, you got scale one. All right, in this case, every variable have the same scale, and this can help us, you know, to assumption that they often. 啊，好，这样，这个，这个，这个，大。

[43:57] 发言人 1: And for sure, it’s because I hate his estimate, so it does not very clear to cover the true desk. Instead, after observation, can introduce some most

local estimates. And averaging over possible is has give us like expectation of density function speed up like integration of the kernel multiply by uh the view which is the probability.

[44:27] 发言人 1: And this is a convolution right this is a definition of convolution if you know what is um what the convolution this is the definition of convolution in mathematics. And again，I call this whole thing，we are in mention like about the age，or that what we are in superization，but this is a。somewhere in text so because you have to just nearby alterations shop structures may change with.

[45:00] 发言人 1: A larger band that you have the age as larger right the asking it becomes smaller beats on the house lower balance become two and that is means this part right if you have a larger balance here in the field right. Um，and if you have a smaller than that，more sensitive，we be examples。And there is also a bias and various tradeoffs in the estimation.

[45:33] 发言人 1: So maybe it will come from two sources. One is smoking everyone gathering something mice. So for small um family with age, a lower smoking bias, it gives us a lower smoking bias and also a higher values. Right? Higher values means like the guys more on.

[46:00] 发言人 1: More like easy as we said, like higher and network of larger bandwidth, um, which means age is larger, we have a higher smoking bias, we have lower back, so a good bandwidth balances this two effects. So and also the sample size is very important for um current business estimation. Right, if you have, for example, don’t have one fine, right, you only have one fine, a single caution.

[46:31] 发言人 1: Right, so more operation reduces reduce some the running function fluctuations. So if you have the answer larger, that means you have more reliable local estimates. I’m increasing number of samples about smoking because um the age or the amount with is a function that um controls the smoking think about like basically have more chances they over.

[47:00] 发言人 1: Distribution of points will be the same as like the distribution of points of less points, right? So for example, there’s for example, we firstly have n points, which are distributed as a like a better distribution with alpha and beta, right? And then you have two n points.

[47:22] 发言人 1: Basically gonna form the same kind of distribution and like just double each point by two right and then in this case we do the estimates the two um declining points gonna give you like the same estimate of the kde contribution because each of them have one or two and as their weight are the other same if you multiply one or two by two you gonna be. One of her hand right it’s the same on the sun oh if you have that samples. And your volatility requires choosing both M and H and its number of sample or the sample size and H is a random variable.

[48:05] 发言人 1: And we can also do like the as we said local estimation in higher dimensions in two dimensions and they were looking uh with bread h has a volume h to the power of d and the access local information depends on multiply by h uh to d and as dimension increases the h d also shrinks. Right，if you have more dimensions，this can be smaller and smaller and smaller，and therefore local

region may contain very few observations。So if you have low dimension，it’s fine to have like um um，you know，you can find local volume is harder，so it’s it’s easier，or for higher dimension，local volume is very small，that means you have more data，right？Um，Assuming happen for a possible machine。

[49:00] 发言人 1: One you know they start of curse of dimensionality curse of dimension the best thing is you have ten points this ten points are in one dimension you have a bad understanding of the point but if you have a higher dimension view of this point for instance you have like three d. Uh, three space three dimensional space which only have ten points right? They can be very separated from each other.

[49:28] 发言人 1: This is a curse of dimension so it’s also happening here right? If you have a high dimension for sure you’re gonna have more points do the estimation instead of less. And that will plant with is a model choice.

[49:43] 发言人 1: Um，The practical procedure for KDD is，I firstly choose several candidates，um，candidates，and then basically use some training data，and now we actually on validation data，and then select the best back back，using validation，cross validation，validation operation。Calculate the loss by whatever and must be a national block each which is love black people. All right, we maximize the black people of happy happy happy validation.

[50:18] 发言人 1: ok. And just which one which experience best 呃那个那个那个结果 maximum maximum like. Um, we still have few slides, but I think we have to take a ten minutes break before I continue. So like we’re gonna start, please have an impartial intention to this talk and ask me, alright?

[51:12] 发言人 1: 我扫了没打开。

[51:31] 发言人 1: 就是。然后。

[51:52] 发言人 1: 我。

[52:05] 发言人 1: 包括这个。

[52:19] 发言人 1: 好的，是。他自己的时候，他是 MVP。

[52:30] 发言人 1: 这个。不是。你。啊，怎么。啊，好吃。

[52:47] 发言人 1: 按照全包那个。对吧，像这个，你往上推了。啊，外界。

[53:31] 发言人 1: 好的时候。

[53:45] 发言人 1: 来帮你找。

[53:56] 发言人 1: 怎么说呢?

[54:13] 发言人 1: 但是就是说嗯你先知道这个我们看这个。等一下啊。但你包含自己是个大小，对吧？你在这个情况下，你有一些时间，你可能有些时间可以给你一些额外的信息，给你去信息搭配的这个开发环境。上面。所以。

[54:41] 发言人 1: 非常谢谢你的初心。让他们现在增长最多。信息调整。然后。确定确定。

[54:57] 发言人 1: 就是这部分就来讲，说为什么它是比较好的，二就是它其实已经逐渐的向。他把这样的一个参数加上这个什么失败的系数，更新这个再给这个数据能够派的这个这样估计。所以我们在这个情况下。放心吧。但这个是那种鲜艳的，然后就是不会再去派派给我，你

成立在。我就看你的情况，你刚才代表什么？你说的有信念，有这个信念之后。比如说打榜的就估计看时代大概是有多。

[55:35] 发言人 1: 对，然后你的实验，对，这个实验是成功了一次，失败了五次，对吧？总数是变成 87。八八七五。就是。

[55:53] 发言人 1: 这就相当于我们这个鲜艳的。我那是屁用，这个我自己。然后那个这个实验的话，就是从。希望。就可以有十万左右和四。那这个。

[56:15] 发言人 1: 放心吧。鲜艳的鲜艳就是假设。

[56:28] 发言人 1: 就是我们先处理一些，就是。

[57:00] 发言人 1: 这个合同跟实验，或是反正二三项这个实验，对吧？但是二三项这个实验里面对你概率比较强一些。为什么整个家庭应该还比较好去。他们正好那个那个。新的。

[57:30] 发言人 1: 你看这个方式。嗯。

[57:39] 发言人 1: 对，那就是说我们刚开始用的还是这样。我。看比如说变成这样的。同样的，或者说权利联系。这个项目。在。哎，我。我们刚才。

[58:04] 发言人 1: 那这样的话，这个效果。相信更加可信，我们应该相信，就是这十天为止，你还有什么问题？我们是 820 年。预期啊，不确定性的东西。什么。一起。

[58:48] 发言人 1: 有知道我们这边是什么？在这个情况下，还没给。这月就是这个曲线这个对性啊后者是。还是。那我们得到是。你是真的有。啊他的这个均值。那个颜色可以选。嗯。嗯。

[59:24] 发言人 1: 啊。产品。

[59:46] 发言人 1: 然后位置啊，就把这个。嗯，东西是。就方便。为什么会发生这个？怎么去创造这个。

[60:13] 发言人 1: 也就是这三个。

[60:20] 发言人 1: 继续。可以，咱们的小超，小超总。不会。就是说啊，咱们。

[62:03] 发言人 1: 学习。

[62:17] 发言人 1: Alright，that’s some，that’s extra lecture。

[62:31] 发言人 1: So we we we mentioned like um how are you gonna choose an age which is not good alright and then uh we also like mentioned this gonna be chosen by um the validation data and so um the training data so here’s the reason why we not gonna use the training data for like people. Because the training point always produces its own karma, right? When h becoming zero or approaching approaching to zero, you.

[63:00] 发言人 1: This distribution or this contribution becomes extremely large, and all can pop some high shrinkage correctly of very narrow spikes along observed points. Therefore, training fit is not a good since the answer to that is in the voice this problem by that is an office and operations. Some practical operations you should be aware of, you know, you can be as the first one is to yourself, as you said, right? This tends to turn in neighborhoods, so you must be to make sure right? Every dimension they almost have the same stuff, right? It is normalized and supporting you, you said.

[63:47] 发言人 1: 第二个是荣耀 X，right? 那些 kernel may assign，let’s say outside the valid range，right? If you have a Gaussian，you have a specific range，for example，like that can cannot be larger than zero。Then the calculation may give you some values even smaller than that’s the problem that you have to solve right the third is numeracy and those we have very small positive price computation of all the components right so this are like problems they have to think about especially in the implementation.

[64:23] 发言人 1: 要 offer. So let’s go summer for a. Um，Can you estimate us by averaging all the contributions? Um，where we are in half like the DHS has because one of our has to the power of D，being some dimension of the effect of X，and then multiply the summation of any like the kernels，each kernel is a function for。

[64:51] 发言人 1: 啊，each observation，then K X minus X over H is corresponding carbon rights of X on，so each operation. Sorry，my main idea is each optimization introduces more local density by using a pay option currently。Second，好，one weapon controls resolution，right?

[65:13] 发言人 1: Second，you have higher age from a small beach，wouldn’t change，right? Third，more dimensions you have more data，this is related to what we said about on the first of dimension，and finally，and what selection is essential，you have to select the。本来是啊。Using like opportunity right.

[65:38] 发言人 1: Maybe your place is a fix um local justice chain with um the data given local estimates. This is a power of as a. 嗯，before also mention，I just want to quickly mention that for next time，it will take two hours or one hours in the maintenance。

[66:09] 发言人 1: Um，and the remaining time is for for um class time，maybe I can introduce more than machine learning before I start the basic and classifiers，because this will require more time to be talking about，so this is some resistant of time，um。And as we said, you are allowed to use one chicken chip, right? Chicken chip, you can write whatever you want on it.

[66:39] 发言人 1: That will be one page, right? Both sides of one page. And you are also allowed to use calculators.

[66:49] 发言人 1: 啊。You can check which actually don’t have a red strip. Like criterion for the reason of the. calculator，Like which calculator you can use，I think there might be one for all twenty times，so if you have to like check on that，you know，like they，um，you know，there will be a website，obviously，you like，you might provide，I think，just provide the link。呃，对，哦，嗯，both conquest and research，I can find。

[67:32] 发言人 1: 速度。Yeah, so this is a list of calculators for examination. For make turn is fine, alright? For make turn is fine, but the final exam you have to, if you want to bring a calculator, you have to make sure your calculator is in this list.

[67:54] 发言人 1: Alright，this is a list of a group calculators for examination。I think this is a simple。Or and for all the process you have, I see in this outcome from the one.

[68:04] 发言人 1: All right, if you have like your companies outside the list of this document, it’s not a lot. All right, it’s not a lot. For me, it’s fine.

[68:16] 发言人 1: I don’t have any restrictions. I just want to make sure I can understand it. I want some.

[68:21] 发言人 1: Maybe I can also do things with him. 嗯，谁好难受啊？嗯，可能三。

[68:50] 发言人 1: 是。Like you don’t have a list for which algorithm can use and which cannot. I was just like, oh, we need here and it’s not for mix up.

[69:00] 发言人 1: All right, it’s not for mix up. Mix up, just add it. Or anything just attached to your calculator, or you can check which one is the cheapest on the market.

[69:16] 发言人 1: Act my ideas are like stupid, like can ask for something to do PR on PC or go off right? Actually, I bought a calculator when I was a fresh man in other classes at City University, and it costs me like five hundred. I think it’s not that expensive now.

[69:37] 发言人 1: If you, if you, you know, just check which one is the cheapest, or you can even buy it on 咸鱼, right? You can find on here somewhere, you know, just like sell very cheap ones. Anyways, maybe I like save for you to save money.

[69:54] 发言人 1: 啊，这三色来啊，明显 classroom。so. Very normal density right for normal density estimation. So, then we are thinking of like if we have the density or the density model, density models where where the density happen like the gradient going to zero. And if it’s a part of peaks of those density in one dimensional cases.

[70:23] 发言人 1: If you see the figures here, you can easily see like these are like the peak of the densities, right? These are the peak of the densities, and as easy or as actually running normal to assume maybe these are like different clusters, and if you see the values here, right? They actually form three different clusters. So, so these are like the intuition why we want to use the density as saying.

[70:55] 发言人 1: So this idea is to have a starting point, and then we try to find a higher destination, and then we can have a local. So points are that are converted to the same or are assigned to the same cluster. And this is not the same as k means because the cluster centers are not initialized directly.

[71:21] 发言人 1: Production kernel, um, if you ignore constants and not parameter packs, um, you gonna have to keep have get has proportional to the exponential product because like. Very higher to us one one or two five uh sigma on the right right and then define the of the w i x because um this and then the p x p x has this in proportion to the nation of this which is which is like near operations should see on larger rates right because. Be closer to like some of the clients I have are.

[72:05] 发言人 1: And for one function waits um we have this and it’s great and take the derivative of it with respect to x. We’re going to have this, um, personally, um, itself because, um, like the final, like the, the growth of the x is the x itself, right? So using the channel, we’re going to, so, uh, the other i have multiply by one over, next is one over two a squares multiply by two, uh, 你试一下。

[72:58] 发言人 1: and therefore. Uh，the next step is，um，you can basically say, well，because like this is a single term，right? And totally have，um，and a different term，right? Maybe someone，let someone is there，because，um，the like derivative of function A plus B equals derivative of function A plus derivative of function B，so you have，the derivative of B X has，is also proportional to one, where it’s square multiplied by this，uh，from here。

[73:28] 发言人 1: Mostly waiting points for observation with large local widths. All right, this is what we call the local widths, the W I package. If I as this, all right, if you’re closer, the one of the observation, you’re gonna have this one of the wide, right? Even if you are like some points. UM, THEY ARE FASTER TOGETHER OR THEY ARE CLOSER TO EACH OTHER THEN FOR THE X CLOSER TO THIS REGION YOU CAN ALSO HAVE LARGER.

[74:00] 发言人 1: Uh，I know，because more x minus x are x i are right。And the way displacement which is versus displacement which is uh the what you call like the gradient and the arrangement like this one can do right like this right we can um put this part out and yeah also fine they um yeah just want to find this part like the sound of waits and like let’s just um you know like violin. Minus x divided by this part has what we call mx mx is the proportion of xi over the full part right which is what we call mx mx in right right right like waiting.

[74:53] 发言人 1: The axes to all this points，all right，well，it’s an axes to all this points，and we can never. Points from x words saying the the mx for the local with in so for an option can enjoy the life like this and after all the possible we have one of the square most like this and this heart inside the practice as a displacement from the current points of local version in which this work on assets. 啊，As starting from x zero with repeats.

[75:36] 发言人 1: So I firstly calculate the weight and then compute the new acts. And this are recomputed after each iteration. And iteration stop when the acts is exactly one.

[75:52] 发言人 1: This is a dominion after and can see it’s almost it’s really similar to what we learned before. 他的刀。Make some model or any each right. What you have, what you have usual points, I mean like you towards something specific points, and then you are like convert and distribute other results.

[76:14] 发言人 1: Consider one dimension all relations x one equals zero one equals three equals two one equals three equals four and start with x equals one h equals two h equals one equals eight. Right，and for two is the same，and for three is larger，right? It’s larger because for F three equals four，it has a lot of ways basically。Sorry，it’s actually like further than the other points to the X zero。

[76:57] 发言人 1: Right，the distance from4 to1。It’s three, right? It’s three.

[77:02] 发言人 1: So square of three is nine. Nine over eight offer this two. The distance is only one.

[77:08] 发言人 1: So you have the square of one is also right. This is right here. And then like you have the three values to be up. Although the different first to have a

higher rate on the last one is smaller. nearby observations contribute more strongly to the outputs.

[77:27] 发言人 1: Using the previous ways, once you have all the weight, how do I said you can calculate the real center or the new um x which x x one here become so the weight multiplied by the value over the sum of the weights. Alright. And it’s like one point four seven right and then my most two one point four seven as a new location and then you can come to the voice again right you can come to the voice by the location of the three point.

[78:07] 发言人 1: This is a like procedure of membership and different sign points may conversion the same boat. Like I can give you like several samples here. You see, for this one, if you are utilizing here or here.

[78:26] 发言人 1: The conversion same up and make the black star here. Also for this, you have this road and this road, you’re gonna go over the same. But if you go a little bit check out out right here, it might converge to a day in December here.

[78:41] 发言人 1: All right, what is going on here? 实际上来看。Uh, the thing for a membership, and I will also appreciate if you want to like play more with it, um, like constructing data set, uh, using for example server calculations, and then you can also try some. To use and plus the intersections like why we care right to see how the relationship converts.

[79:13] 发言人 1: Several things here like find the weather controls that on discover bolts for me ship does not require a fixed number of clusters and instead of that controls the landscape for smaller age of man and now does the piece happen then it gives us more clusters but you can have a larger age and you might be submerged when you get fewer clusters.

[79:40] 发言人 1: Alright, number of clusters is not defined by you, many some on by us。Is um defined or is actually um derived by different age. All right, the age is set for like music and like um starting talking faster like happening in. So that’s um time with controls of smoking and um classroom structure.

[80:10] 发言人 1: And compared to major with techniques like this matters on they they also they both repeatedly compute means the techniques that they use hard plus for membership this is also like the the the most criticism coming from the techniques right even like we talk about gaussian mixture model in the same about it’s um it’s using a soft membership. Compare things so, anything is hard, faster membership just gives up, and then use this, you like the fastening, right? This is calculate the ways of them, you know, like this, you know, the faster.

[80:46] 发言人 1: And we should use the place that the talented ways, and then to get the local ways at you. And keming says like the kas specify directly yes how we know how we send how. How many clusters you want? But for me, said, um, makes us very emerge, um, as the most, it really depends on how many models we have, and number of models are divided by them, divided by the set, in the age of set.

[81:15] 发言人 1: All right, so that must replaces a K. As a man is gallery or the man is stop. I know also need to consider computational cost a direct mentioned after evaluates distances from each location each location to all the observations from n

starting points in the dimensions of the series you can get this all and square dt and for larger datasets common approximations include neighborhood searching.

[81:48] 发言人 1: Or how has for purple shop，about some point you are starting patients，this are different approximations for the large data set，because we have lot of this in element。The um as where in that increase exponential right um. 所以这个意思呢，可以 potential more expensive campaigns，because campaigns other hires，I think is all and。I think so, and also not that something like that.

[82:19] 发言人 1: It’s definitely not this way. All right, so this is also something that you have to know if you want to work hard on computer science after graduation. For each algorithm, you need to consider the computational cost. If you are often you get a bad result, it’s not the end of the day, it’s not the end of the story.

[82:42] 发言人 1: Alright，you have to also get comparable computational cost or less computational cost。This is also how they，um，more than machine learning or deep learning develops，right? You want to have the same，um，you know，you want，you want to get it，like as you result as I。

[83:08] 发言人 1: 嗯，这样子在这儿，其实没什么想。Uh，and there’s also some practical common choices or initiative，Uh，initiative results，and somehow on Zhang Jiao，because face，important choices include，Uh，biggest family，back west，and also volatility，marketing tolerance，and soon you are going to learn all, Uh，most may emerge in one time across。

[83:43] 发言人 1: 好，finally，I want to introduce another argument called specialty classroom。

[84:00] 发言人 1: Um, this this method is work well for compact groups of a lot centers. Or if you have like, um, you know like, this shape or um, spherical shape clusters, this basic clustering methods are already enough. But you can struggle with um curved structures where two lines making far apart uh super connected to a long neural relations um so special classroom uses uh with connectivity instead of distance to one sentences.

[84:38] 发言人 1: Each operation x i becomes one graph model this this is some information that you might learn from discrete mathematics right so we have constructed graph graph means like you have a lot of points there are each set of points gonna be one of the most in the graph and this set of points can be connected to each other by edges. And the age here is the servant like the similarities of this data points. So, as we right here, nearby notes are connected with similar weights.

[85:13] 发言人 1: Right, you have, for example, W I J, right, equal to zero. This is the restriction for all the weights. And how many choices?

[85:34] 发言人 1: 啊。啊，比如说我拿这个 H 去看了，which consumes critically similar to the previous distance。啊 larger as or smaller as can serve as their functions. And we connect all ways into a larger measures we have the the average as say uh the measures of all the larger i write case a larger i write in the smoke. connected.

[86:04] 发言人 1: And then the similarity tells us like each individual connection of our most high is total connection strength is like the relationship of this point to all the

data points in the data set which means we can it basically sign as the summation of all the all ways from this time to average day except the it is up after it. contents is off which at one more today on i will be on so connect those values in the grading matrix um. We had a T to be on final T one to T n and that’s the W scores pairwise similarity while the D scores how strongly these knowledge connected overall right.

[86:56] 发言人 1: 呃，suppose you know receiving us their 呃 values，还有。For all I compare its value with its neighbors. Alright，right，compare each note，this inverse thing，right，would have，um，so the receive ability is a value that we see, which is the data，all right，the world data，you are really have like the connection, then think about you，you get some different data to this。To this model, what’s gonna happen right? Promote I um parents that is with his neighbors and gave us like this nation of that’s what I think which is um this strength between a low I and low K also try by f i minus f j because we are comparing. Every year you know for this two different months. And expanding it gave us this going to TI I find which is submission.

[88:00] 发言人 1: Multiply by i five minus submission over j，um，have the ij above fj, all right，have the ij because this is not work，have the i again here，so it’s not for i i integer，so we can’t write it like simply。And it makes it small，they give us a B minus W positive by a pack，and this alleviates the graph partition，which is called B minus F。而就是 demand time，他有 back matrix，back matrix，with um，it’s um，back matrix，as estimation of each line of the um，the weight matrix。And for all time, the L F equals to, um, as as we just, um, right here, right? so if.

[89:00] 发言人 1: The neighbors the constant is small and it is it strongly connects it most with a very different methods to become one and usually out measures the variation variation related to relative to local neighbors. But not local neighbors. This is useful because finding inside the same cluster two baroreceptors over strong drug connections.

[89:27] 发言人 1: Right? This is this kind of time because once you have a point, if that is, um, uh, like the value of it is very similar to like this roughly connected neighbors, you’re gonna have a way smaller disaster, right? Because the high point has high value.

[89:44] 发言人 1: Oh, you may have a large outlier, right? If you have a, um, there are different values for F and FJ, even they, because they also have where they have really high outlier, and this is. So the four four ground variation is uh f t r f uh which is one hundred two multiplied by the summation of.

[90:16] 发言人 1: environment is at this square and strongly connected nodes have one life arriving so assigning them very different values is heavily analyzed. Now basically means you wanna have that if the points are also similar to each other we can assign them with um similar ways or similar um uh f five or f f values right so small value of f therefore means. 啊，strongly young connects the cultures in similar values，and this is feeling。Okay，so after you know this，I know you think about how you.

[91:00] 发言人 1: Copy graph，right? One way copy graph，supposed to graph is divided into two groups，P and E，and the top with crossing application，crossing application is the copy A B，which is the final summation of W I J，where I is from A and J is from B，we are summing up all the base methods。From each one in eyes and each one in day and useful question should have strong connections with angles and weak connections to triangles.

[91:32] 发言人 1: So therefore, a small heart value. The washing gives some matrices or communication of the separate separation of text. We’re gonna see it in it.

[91:46] 发言人 1: So, in color super communication by um I five plus one or zero, one is strong and zero point d. We’re mostly the same group, you’re gonna have I five minus I five square equal zero, right? Either you’re gonna be like one.

[92:05] 发言人 1: I got from different groups they are the square of the this the difference in one and there will be a pushing energy concept the way to crossing the occasion therefore like the fclf here means they are the half ap right this is easy to understand because we only consider.

[92:26] 发言人 1: The ways only if you are in different groups for top a right that same here i iphone a and j from b so defining i buy like this living process i see out as is a is a color baby the one costume can be express for the fashion right which is just fashion and then we can we can do whatever make sure some patient that we we need for this fashion. So minimizing only hot a be an astroid. 啊，新高。Weekly connecting posts defining graph following as the summation of the eyes. You know, the eyes are they are the right for each eyes, the sum of the corresponding path growth of the nation of the great nations. And it’s the the strength of point at point eyes, all points in the dataset.

[93:24] 发言人 1: Alright，it’s making of the ways，and not like hot balance separation and group size，so it’s like car a b over volume a plus car a b over volume b，to make sure it’s not，um，yes，it，it is discovery discovery。那个这个时间可能，this is actually i think i share the paper onannouncement before maybe last time，或者那个那个那个那个那个 long life，happy right，this is because one paper，i know。I share like two weeks ago, one two weeks ago, one week ago, right? And you can see the the things like we wanna we wanna estimate we wanna you know separate it by normalizing the car using the volume and use the volume of and to make sure it’s have normalized otherwise.

[94:33] 发言人 1: And from broadcast to inatters, one want to make this costum with like their health fund searching over all this is great qualifications is very difficult and relax them is very faster in there for a little manufacturer as some as some that something want to right the number is hard the relax the problem becomes. FCL and over FCTF where F is for the technical team. 嗯。

[95:08] 发言人 1: so as a historian the new line here is like. Uh，so stationary solutions in case finds the L F equals uh long idea，and the classroom problem therefore converts into a eigenvalue problem，which is the prime eigenvalue of the Navier Stokes out，right? So because every row of outside zero, so we have L one equals zero.

[95:42] 发言人 1: And um, I want this like yeah, just um, you know, have so zero here is also vector, so I here is like matrix and one here is offline vector, and you do like matrix multiplication, if it better to do some of the show, you can have here. and smallest cable in that view as their one of the line of zero as a cost making action. right and the cost of bad not separate notes the the even better associated with the second smallest impact is the first function resolution and she is slowly across on strong connection with more strong uh strong way across good connection.

[96:26] 发言人 1: So for two clusters this eigenvector reveals the next graph separation. All right, so for multiple clusters we’re gonna use like several eigenvectors associated with the smallest smallest eigenvectors. Matrix the wrong eye of you becomes a new representation of the operation of I so that you this in the new idea to represent like the original exxon the new product is described like.

[97:00] 发言人 1: Graph connectivity rather than the original Euclidean geometry space. Right, we are comparing the distances as the center by this new value that tells us the relationship between data points or the nearest neighbors. The current tools have become easier to separate here on spectral space.

[97:26] 发言人 1: So it is contrasting into like um pipeline. Um, it’s like first way we build a local similarity graph, and then call the similarity matrix. With um awesome format, we can compute the degree matrix, which is the similarity of each row of it, and put it on the document.

[97:46] 发言人 1: And then call the graph matching, which is the D minus W. And then compares the low agent value agents factors and use their roles as special features and what paintings on this features. So this is.

[98:07] 发言人 1: So this is a you can see this variation or variation of it means by the differences it means use the feature the raw features and here we use the spectral features constructed by the raw。So transformation here is we have the original data construct local block and then we use spectral clustering or spectral implementation then do and this cluster. 嗯。

[98:40] 发言人 1: The ground construction is really important part in the construction because it determines how the features gonna get right. The special clustering has properly on the similarity ground experience. Important choices include how many neighbors are gonna involve for computing the the the K D alright.

[98:59] 发言人 1: And then the. Similarity scale has, if you have a smaller scale, I can control the exponential part of it. If you’re using a fashion partner, and number of clusters k, all right, how many numbers you want on the clusters? Because now at the end of the day, we’re using k means, you sell the need check or some other.

[99:20] 发言人 1: UM, YOU LIKE METHODS? SO I HAVE TO FIND THE NUMBER OF CLASSES MYSELF. SO IF A GRAPH IS TWO SPOTS, ONE SHOULD GROUP MAYBE MAYBE HALF UM DISCONNECTED, AND IF A GRAPH IS TWO DANCE DIFFERENT GROUPS MAY BE COMING STRUCTURAL LINKS.

[99:37] 发言人 1: THEREFORE, GRAPH CONSTRUCTING IS PART OF THE CLASSROOM MODEL. 啥叫 C M 都不相信是。And you definitely consider about the communication cost before you connect similar nature super profit and square pair

of similarities because you have to to confuse that similarities between each pair of the data. But data the the the aim of master computation also has essential costs. Uh, or cost, uh, for large, um, data sets, common approaches to, um, you know, reduce computing costs, both stars nearest neighbor was, and this one not constructing the nearest neighbor was, for all the points, but only consider the nearest kind of timing。

[100:24] 发言人 1: 啊，thereabouts right, for each, for each acts, and the approximate neighbor search, across we can use it like by using some tree structure, like a tree or some other tree structures, or we can use like sparse even solvers, instead of like solving even getting the big things, and like we can use new approach, of constant errors, to get the big values and big big matrices。And also some of all that market methods the second class is usually more expensive than kab because kab is only part of it. Before that, also need to construct a graph, which can take an square of conversation cost or square, but can represent more rich of like much information, in the aus geometry, because they are not only considered the distance to the center, but the distance to nearby data points。Alright, and then I want to compare and summarize, um, you know the methods we have learned for four different, um, passion methods, anything, GMF, each respect the question, so the this like the definition for passion are different。The pennies are there really points near the same central of the jama.

[101:51] 发言人 1: We are saying like the points the points in the same class are easy to explain by the same uh publishing component for me. Each uh always conversion the same as the model and perspective classing and strongly connecting in the same one so guys are right their definitions are different and among for methods. Online donation uses a bandwidth to choose number of faster another three is on。And simple limitation for this port are like this or any is because model is so simple.

[102:32] 发言人 1: It’s a simple tradition among this port that is only have one you know one primes that this thing actually poses really strong restriction or assumption on the data set. So it can only work on compact function or any noncompact function in the file. Right, 啊，不是 GMM，啊，它还是 distribution something like adoption, right, where the induction that seems to have.

[103:00] 发言人 1: All data are the submission of different patients. This is a assumption here that links the performance of the hand. There is not possible has a little market or beautiful tissue building.

[103:15] 发言人 1: And something to work. For me, shit, it’s about less sensitivity, respectful class, most marketing is competition cost, because it’s all in square, larger, right, just a. 哦，they also have the problem of that last sensitivity, but with some, you know, smaller compared to conversation cost。

[103:41] 发言人 1: Assume that there is different factors because each method defines faster structure different. This is like how the machine learning work because machine always have this sort of problems. um for.

[104:00] 发言人 1: From global models to local structures. So here I want to say that the methods for a perfection how structures are presented. So central choice, no way

forget the probabilistic distribution, and then we talk, we think about like best involvement, and then we’re often connected. This crowd comes also like Chinese GM.

[104:24] 发言人 1: But initiative and then special costum, right? And just with with us on last review, maybe as made global distance from nearby observations and initiative also because you could talk let’s work on what the most special costum finally since local cloud connections will be of not in contact with in the data set. And this is some of course, first thing local methods of why forcing all there is one base global shape.

[104:54] 发言人 1: Just like it’s not much can be forecasting nature that has a stronger. Assumption on the data set itself, data distribution itself. Second, that one run with determines the resolution of AKD and also the music costuming, right? And also construction determines the geometry used by special costuming, also depends on the scale of expert. Um，finally，positive assumptions on policies，on instruction，on architecture，and different positive assumptions can give you definitely different positive results。And you，you know，that’s a very old vector。

[105:55] 发言人 1: 在旁边在旁边在旁边标注在旁边标注好需要。在。

[106:13] 发言人 1: 我。

[106:34] 发言人 1: 需要在我的皮肤里。

[108:44] 发言人 1: 好。这个是。

[109:08] 发言人 1: 自己。

[110:22] 发言人 1: 有可能。

[110:29] 发言人 1: 中。

[115:17] 发言人 1: 你。

[116:38] 发言人 1: So this is the first question we have solved. We have three operations, and now we use the time for the curve. This particularly has one, does not say the curve can change all the time, right? And if you use this. And then we can also have. After the pH x hat，which is estimated K D E，um，it’s，um，you know, the density function looks like this，and I want to come into the pH for x equals two, for H one，H equals one，H equals two，and also do this for pH one，given H equals one and H equals two。And second and third um validation observation has this x bar equals one and which time where assigns assigns the largest density so this tells us how we can choose the backplate you just think of validation and based on this calculation this one what changing age changes.

[118:28] 发言人 1: 困。

[120:14] 发言人 1: 是不是?

[120:23] 发言人 1: 你。

[121:22] 发言人 1: 这个。

[123:19] 发言人 1: 嗯。好的，嗯。As we’re coming, so for the first question, um, different age equals one, um, we can calculate say, um, the p one hat, p one two hat equals first one third because we have three data points, right?

[123:40] 发言人 1: And using a time kernel for k two, k zero, and k next one, we can calculate. Because fine value right which is. zero one and zero give them two one and zero and next with one just.

[124:00] 发言人 1: The value here, right? Just plotting the value here because it’s equaling the x minus x. So now we are calculating for x equals two. So two minus zero is two and two minus two is zero, two minus three is one, right? That’s the reason why I have like this is k two k zero and next is one.

[124:22] 发言人 1: Right，and then the corresponding like looking into the two zero next one into this function，and that is two as much as one，which is zero，right? If it’s zero，uh，put it into here，zero is one minus zero is one，and if the next one next one’s absolute value is one，which is on to this equation。And one minus absolute value of maximum one is one minus one which is also zero right so then you have like the values of this particles. Talking into the um the the rating here you know have um as zero plus one plus zero over uh must be one by uh one third which is one third right and h equals two the only thing change.

[125:19] 发言人 1: Uh，and that we can change not the only thing，but we can first change the spirit right，as one over six，since now one over three，and then like all the bad spirit can be divided by half，divided by two，sorry，it’s like multiply by half，which is one zero next is zero point five。And since we have k one equals zero, k zero equals one, k next is just point five, we can talk business, the current function can be zero point five.1010, sign this together as um like the upper two can possibly find on one or six, which is on one important, that this is on p two two.

[126:01] 发言人 1: 变成了那些数字化。And now we can do it for H equals one as well. The only difference is they are like the value to be divided or converting values into you, right? And also like the conditions here.

[126:23] 发言人 1: In this case, we got H equals. 啊 sorrystillstillequalsonesorrysorryfor. See what happen. Do I make a mistake here? 啊，不是，H equals one，嗯嗯嗯，P one has one，哦，yeah，这是 calculate in P one，所以 that’s still one one，所以 H equals one，嗯，we have P one has one because one or third，and then，嗯，having it return together，and all three purple values are zero。

[127:00] 发言人 1: Because outside this gonna just have a endpoint of the current option. Next to outside because this is action values to five and one. So all this guys equals zero. You have you want hats.

[127:18] 发言人 1: Uh，one equals zero，or each equals two，you can calculate pixel one pixel，uh，pixel half one，which is one over six，and k multiply by zero point five，and next zero point five plus k next one，next one，you sum up those data, which is one six，but we have forgot to have the uh。那还是。See what happened here.

[127:48] 发言人 1: So if you calculate it for H equals, I will calculate here. Okay. Alright, yes, all. Yeah, I just like this idea. But anyways, here we have for H. H equals one and H equals two, we have P one has two and P two has two equals one third and one fourth. For H equals one and H equals two, we have P one has one and P one has and P two has one equals zero and one or six.

[128:20] 发言人 1: 哦，没算出来，solution for the first two questions，and for the third questions，哦，for x，哦，which is validation data equals one，you want one，and p two one equals zero，one or six respectively，where the second one is larger，right? We use validation data to select。The best time when so second one sliders you can say well we can use age equals to as a of of the background and for the first question the facts of changing the background increasing age makes um.

[129:00] 发言人 1: Current contribution wider in this example, h equals one leaves a zero density cap and x equals one, but h equals two allows contribution from nearby observations to overlap, and thus a larger background produces more smoothing and shares information over a larger neighborhood. This is the French army fact that’s like you are supposed to answer. Um，the second question is about major iterations, so I have like four data points，and use the function in the shape waste with h equals one，we have the current farming that will i have to post some exponential of maximal x minus x minus square root two。And start with x zero equals two we want to like do several iterations.

[129:55] 发言人 1: Right you have a initial client have the clients you have the ocean current. No，then I can give four ways，for the two different，for different points, and then I’ll compute right ten in，and then I can finish next，and I’ll send the next，and waiting，and do this again for x two。And for the promise to our face, I think by which groups this text story is moving so work and may just how they or just how they on their own and report on market value to this decimal places. Alright，I will give you another um six to eight minutes，and then we will stop。

[131:37] 发言人 1: 看。

[134:02] 发言人 1: 啊。

[134:16] 发言人 1: 嗯。

[135:09] 发言人 1: 好。

[135:41] 发言人 1: 啊。

[137:25] 发言人 1: 谁。

[137:40] 发言人 1: Alright, let’s see the answers. So for share as zero equals two, um, we can calculate the correlation. Ways for this order of their points, we are can just access to here, right? And then we can calculate the correlation.

[138:00] 发言人 1: place and using this place, so um then calculate the updates and sensor. You can also see that as two because one has right this way because it’s a close yes, so that the original center or the original financial point. 我去。

[138:31] 发言人 1: Multiplying the ways with the corresponding acts over the the sum of the weights is us1..355，and then we can compute the mean shift factor which is

um1..1..355 minus two，which is maximum zero point six four five。So the first update moves to the left，and from0 to1..355。And second, I have this one hundred and five points.

[139:01] 发言人 1: We can again calculate the different ways. Still the second point is the largest way. And therefore we have the ways to be this for numbers and operations near zero and one dominance local way to me.

[139:18] 发言人 1: So use your use your ways we can calculate the x two which is the the second object second duration of the x. The shares has0..778 is closer to even closer to the back and side, right? To zero.

[139:36] 发言人 1: So the first updates are2 to1..355 and then0..778. The position actually is moving towards the last intensity group near0..1. alright.

[139:51] 发言人 1: And repeated updates with continue reviews for computing work over until patients stabilize here and let’s stay on. This is some hard work on your mission. And the third question you want to exercise on course is on process on broadcasting on function.

[140:15] 发言人 1: Consider with matrix W and compared P one and P two. This two cases, this two on broadcasting. One is compute the degree matrix D first and the broadcasting matrix L which is D minus W. And then compute cost a b for both positions. Positions one is like you like on three four, which is one and two three four.

[140:43] 发言人 1: And then use n cost, which um survives them that the bottom the cost a b over volume a plus the cost a b over volume b, um to um to find the best um position and then. 嗯，to see which one is better，right? And if you have equals one one zero zero，come to attack，yeah，I’ll have an determining way，they’re taking preferred by model。Alright，think about this question，and then。嗯。

[142:14] 发言人 1: 是。

[142:59] 发言人 1: 嗯。

[144:48] 发言人 1: 不是。

[146:23] 发言人 1: 啊。

[148:48] 发言人 1: 嗯。Alright，um，for the first question，um，anyone that name of it，it would be super easy，that you could work，working is B，it’s. Like for each row, right? And I put the number of the nominations today on the entry and some of the entries.

[149:13] 发言人 1: What that is, he wants to be correct before he wants to three four four three right? And then describe the acronym of this key. And then for um L，if you use a D one，just not going to give you a measure like this，this is a matrix，and then the countries can also concentrations，and while they are next of all technology represents，otherwise connections。

[149:45] 发言人 1: Uh，equal sum to uh zero，right equal sum to zero，because of you know construction，because of ways problem points，it’s zero，and if you think this，you can has 啊。那大家都准备了呃操作事。And then I every moment some to zero and now we gonna talk about which patients kind of.

[150:11] 发言人 1: For proposition p one, you have one, two, and three, four s two groups. Only the age between not three and two and three across the process of patient with waste one. So cup p one is some good one. Number p two, the color is one and two three four, and age is between not one and two process of patient. With uh wait three, uh so, um, that is the corresponding uh location, and this can also seen by the waste matrix, right? Um, basically for one to one report, you can see, uh, like there are only uh value one, and also one super report, there's only one value.

[151:05] 发言人 1: 啊。你卡数据光，对，还是 2？那 1 万卡是 much bigger connection。And if you do this normalization for not one one，and some of the constrain values in the that moment，that matrix be the first two values，which is seven，also accepting to connect the second，part of the two values，which also gives us on seven，so normally cup for p one is on two。Over seven, but if you do it for the second one, one one again, two three four seven, so something of this fourteen over eleven, this one. Now it's black around you want again right same as um this without normalization without normalization we have the second one starts. Hospitalization is the same as all years, right?

[152:09] 发言人 1: Um，I don't know why，patient or on one wants，um，results，maybe maybe have the same results for，um，you know，the patient，which means the first patient is bad。

[152:26] 发言人 1: 嗯，they are see out equals one and this equals the like a patient be one right you calculate the cost is also one so about flashing energy therefore measure say similarity wait across in the some private station. right，so this is how the flashing works。Um compared novels that you have approximately zero point zero eight six.

[153:00] 发言人 1: 可能卡第 2 个是 approximately equals one two seven three，all right? And since smaller not less size preferred，so the first one is preferred。This is as we said，it's the same as on top online version。Expectations have this across the way a off way a always one，while preserving school strong pages of range of books。And it's almost like this, our great tutorial, and we are still like around ten thirty minutes.

[153:35] 发言人 1: I want you to have the have this time to ask you questions because you know, like as we mentioned, we really want you to you know have the communication. With me, go the closet. The question can be anything related to the course or the expert or violence sound. So, yeah, this is like model timer.

[154:00] 发言人 1: First，so if you have any questions，you can just，just speak up, whichever you prefer。哎，开心出发。嗯，几个 questions。Anyways, you know what? Like this, I just want to remind you again, right? So for all the files that are inside the context, so if you happen, you know, like you want to find the files we have to work, check the corresponding folder, right? Checks the corresponding folder, I put all the materials inside that, and after each class.

[154:39] 发言人 1: 二月呢嗯 apple 没干对呢，还是呢 apple 那 corresponding solutions。嗯，So they have this as well，for example this course，right so they。This may be

serious one as as here means like solution alright so if you want to see the transformation solution just require to.

[155:14] 发言人 1: first here as we mentioned so I just want just here again. The range of the examination, I think, because I checked the the make some paper roughly, and I believe the the packages at least most part of the Facebook assignment first three weeks, or after three weeks. 啊。And I also like have a kind of reminder on you know just be familiar with distributions like all share like one class share like um.

[156:00] 发言人 1: The song and all the distribution this word distribution and is you know really good understanding of their. Alright, so this is our most important thing。For you and also correction the first assignment is still on the day after tomorrow is not tomorrow is the day after tomorrow night.

[156:29] 发言人 1: I see fourteen submissions for now. I’m a little bit worried for you guys because I just realized that in total um. Uh, see, then posts, I think it’s sixty.

[156:44] 发言人 1: My students are like, I can see twenty students or thirty students at the week. This is a problem because, um, you know, even even attendance is not the part of it. Um, you know, like, um, how is that?

[157:00] 发言人 1: The assessment is not come for any marks, but still attendance is really important. Like if you are not here, you know, you actually have to ask for leave or at least let me know. All right, so this is a problem.

[157:18] 发言人 1: And as I said, I received an email this morning. It shows like at least my students. Have um like um um a temperature less than seventy percent.

[157:32] 发言人 1: Sometimes it’s a problem. I’m not sure how how the converter is going to deal with that. I think it’s a problem at least.

[157:38] 发言人 1: So make sure um this kind of face before you get into the constant methodology. All right. 啊。嗯，还是说，嗯，yeah。Turns out you haven’t submitted yet.

[157:52] 发言人 1: You are in total working some agents. But sports is no permission great because I have to create great after. 啊，you know like national holiday，um, yes，this is the case，this takes fifteen like percent of your final mark，right?

[158:10] 发言人 1: So you should be like very careful about it because otherwise you know you can even you know like cause lose all this points I’m not sure if I make it clear enough for you. But I think the people I have to make clear is like the people get a type of you guys. But anyways, I just hope you guys understand what I’m saying.

[158:34] 发言人 1: Like the first homework take15 percent of the final mark. The midterm is10 percent. Alright, so make sure you submit like the homework on time, otherwise you can lose all the points. FOR AT LEAST THE FIFTEEN POINT, ALL RIGHT, AND. Let me see yeah and again like for late term.

[159:00] 发言人 1: Uh，I think I already showed you what is like the current one approved collaborators。This is only for final exam，like if you just don’t bring your phone，using your own calculator is fine for midterm，alright? You can bring

whatever calculator you want for midterm，for final exam，this refers to the list, alright? I’m going to put it on on announcement。啊，然后。

[159:37] 发言人 1: 嗯，yes again，the change sheet，everyone can point one change sheet，one a four paper，both side，you can write whatever，like。If you can do it like you can even write ten thousand words on it if you can do it you can see some sample i will show you some sample 啊写写。This is a lot for only for maker.

[160:03] 发言人 1: Okay, this is money a lot for maker. Right, you can do it for maker, you can’t do it for our insight. You can even use world like Microsoft world or something like that.

[160:13] 发言人 1: To make this all this works to be small enough. You can see it and then just take all sides. Take all sides is fine.

[160:25] 发言人 1: All right, just make use of it. I was just right here. Otherwise, you can’t really, you know, she can’t. All right, so this is. Yeah, something like this, this would be really useful, right?

[160:39] 发言人 1: You can see how you can, if you want to use the text to do it, you can even use some content and just try to convert most of the information into one like all side of that. Alright，so this is some. Um, you know, like, uh, tests for this.

[161:03] 发言人 1: 嗯，你好似。That’s what I want to say here. Do you have any questions or ideas or things you want to ask? Are you still trying to ask you want to like openly toss some images on it and ask me? Alright, so yeah, thank you.

[161:22] 发言人 1: You can ask questions. If you want, you want to ask questions, and you just want to listen to others asking questions. All right. Let’s see you in two weeks on um after national holiday I love you see you.

[162:18] 发言人 1: 简历不是 C B 吗?

[162:47] 发言人 1: 嗯。

[163:03] 发言人 1: 怎么去。

[163:10] 发言人 1: Beta Beta Sigma Sigma. 为什么好几个人都发过来个 VC?

[163:47] 发言人 1: 市场。

[163:53] 发言人 1: 啊。

[164:06] 发言人 1: 第一个是 E V V。

[164:17] 发言人 1: 我这里改就行了你下午有课吗没课下午有没课没课那我们回去吧好啊嗯。

[164:37] 发言人 1: 其他那个答案是。那我先去拿做饭去啊什么我先去吃做饭去啊等我一下我马上我不是我是去肯德基拿肯德基现在啊拿拿一个就是去肯悦那里拿然后带上来然后去公交车哦好。你先去。

[165:18] 发言人 1: 学然后就像刚才说的那分布说完了。